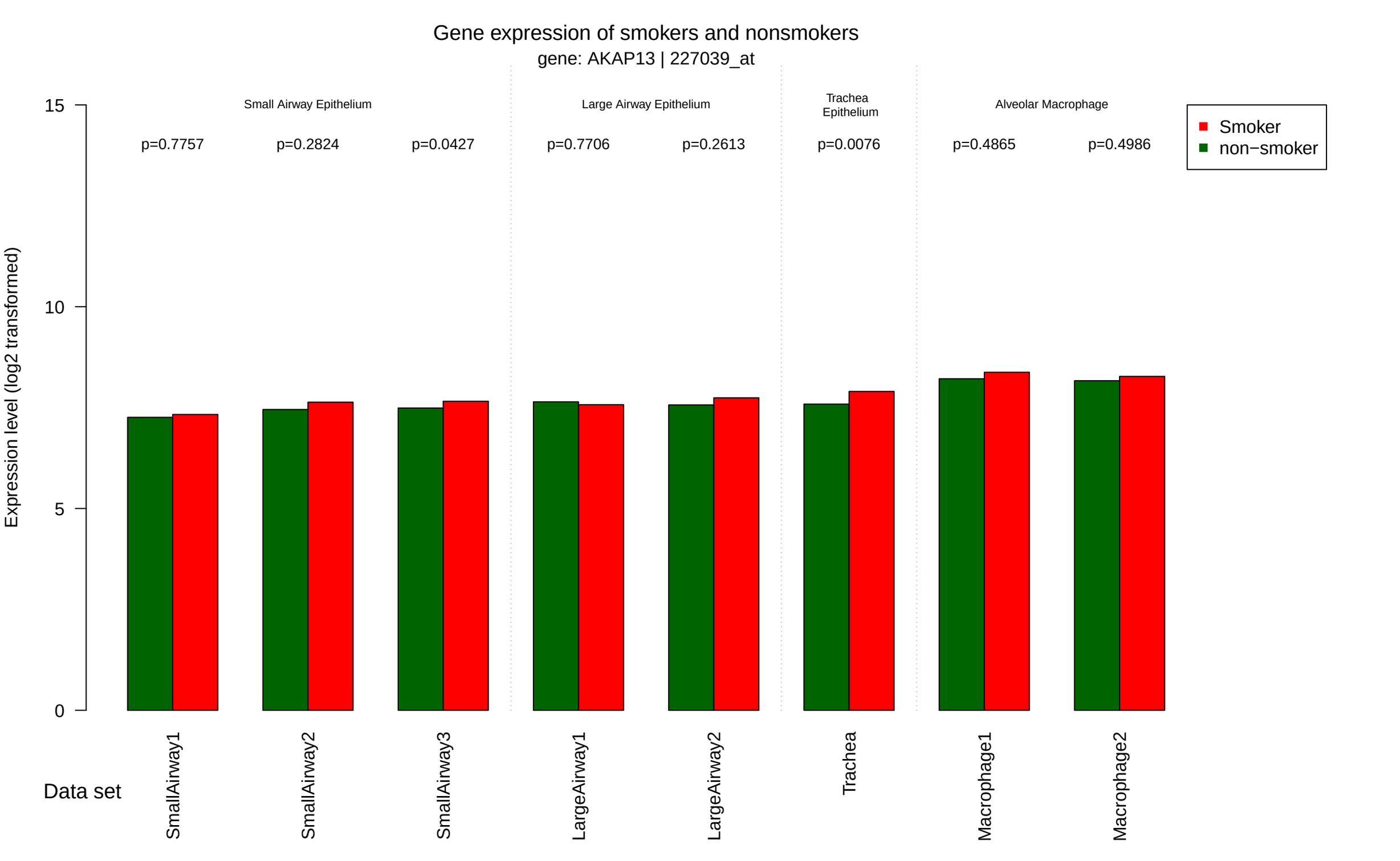
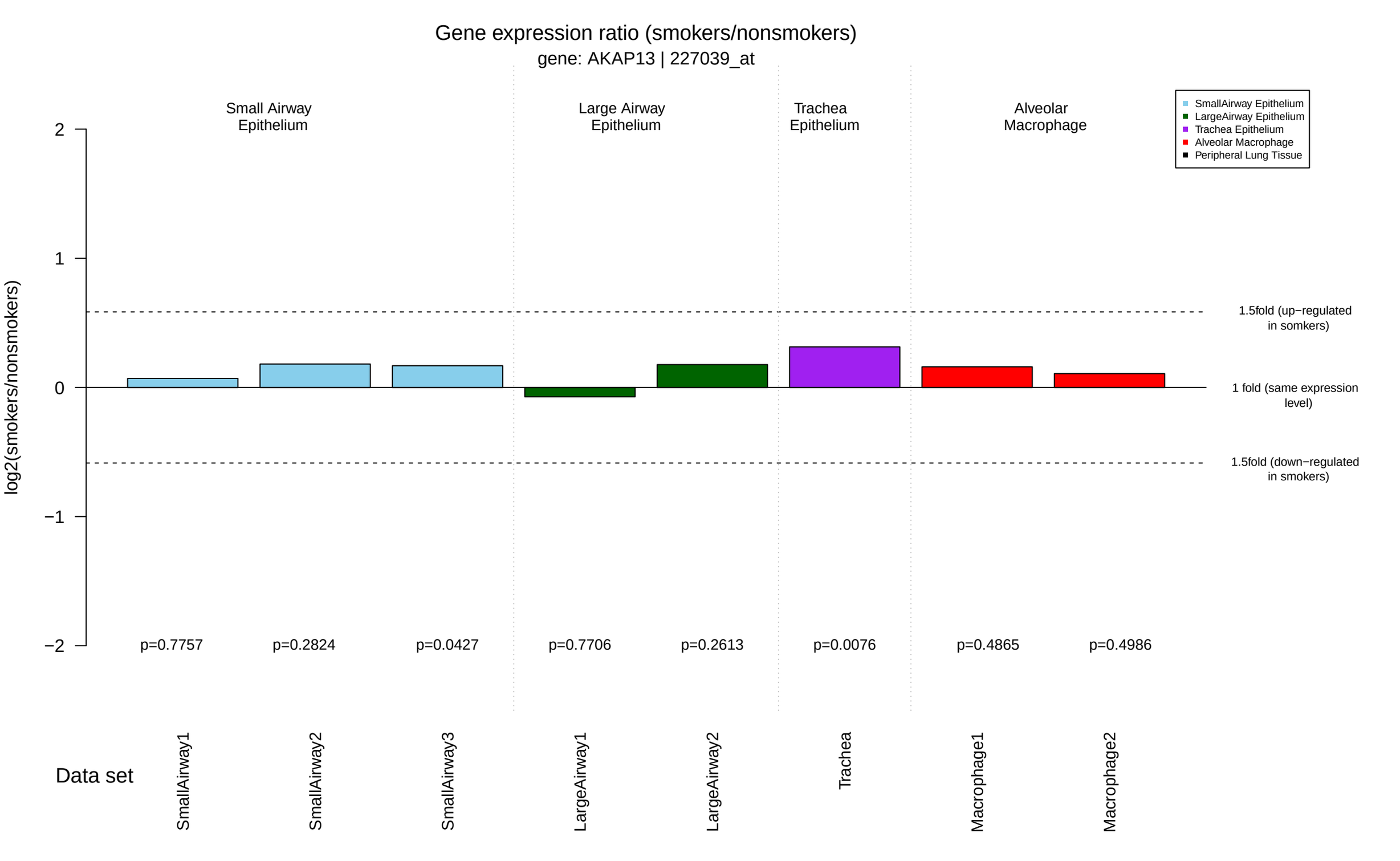
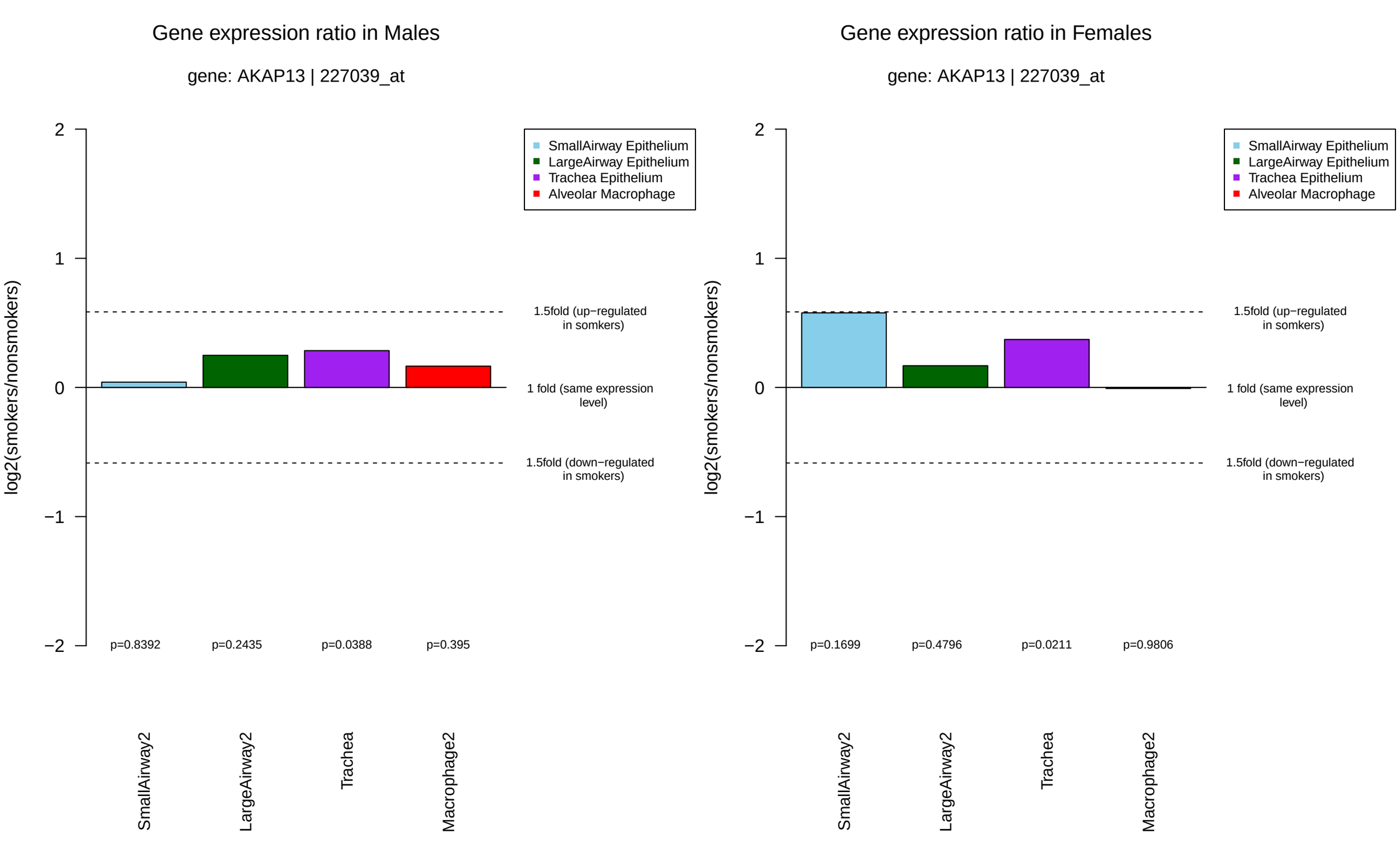
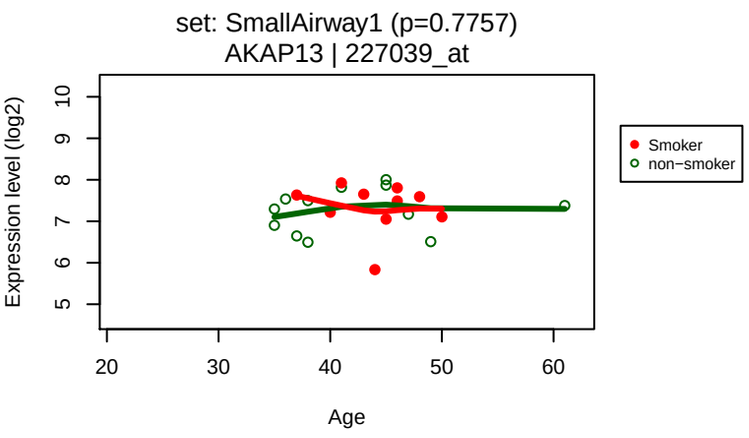
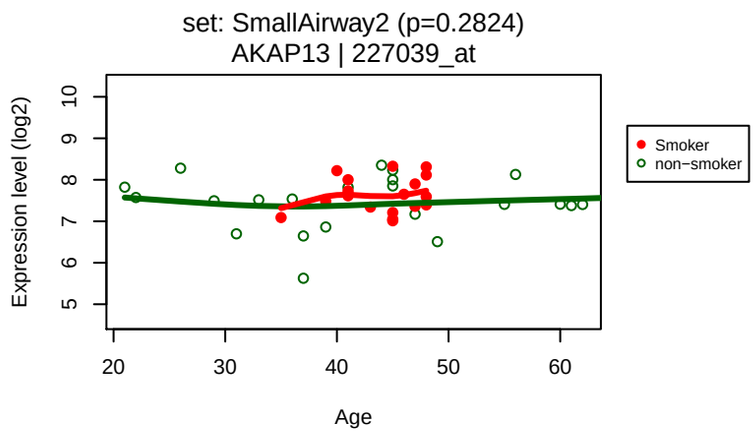
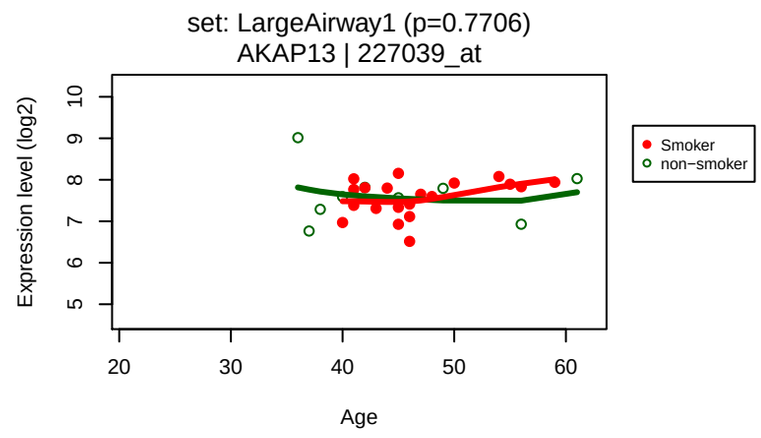

Gene expression of smokers and nonsmokers
gene: AKAP13 | 227039_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7757
p=0.2824
p=0.0427
p=0.7706
p=0.2613
p=0.0076
p=0.4865
p=0.4986
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

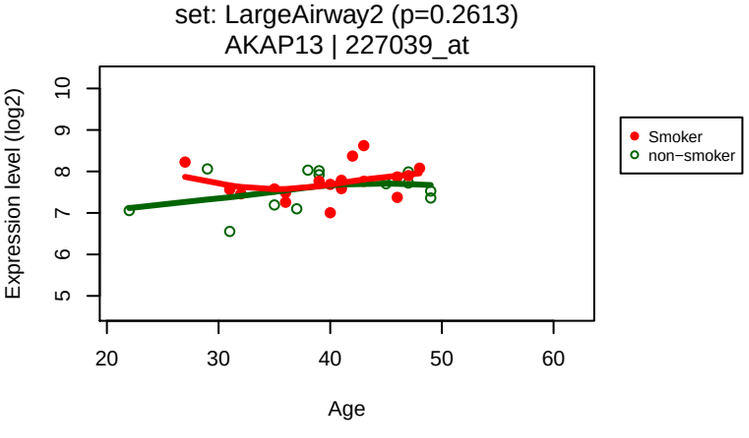

Gene expression ratio (smokers/nonsmokers)
gene: AKAP13 | 227039_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7757
p=0.2824
p=0.0427
p=0.7706
p=0.2613
p=0.0076
p=0.4865
p=0.4986
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

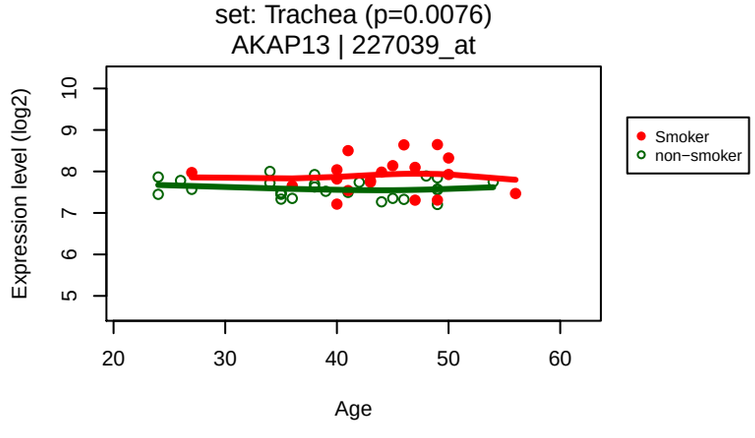

Gene expression ratio in Males
gene: AKAP13 | 227039_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8392
p=0.2435
p=0.0388
p=0.395
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AKAP13 | 227039_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1699
p=0.4796
p=0.0211
p=0.9806
SmallAirway2
LargeAirway2
Trachea
Macrophage2

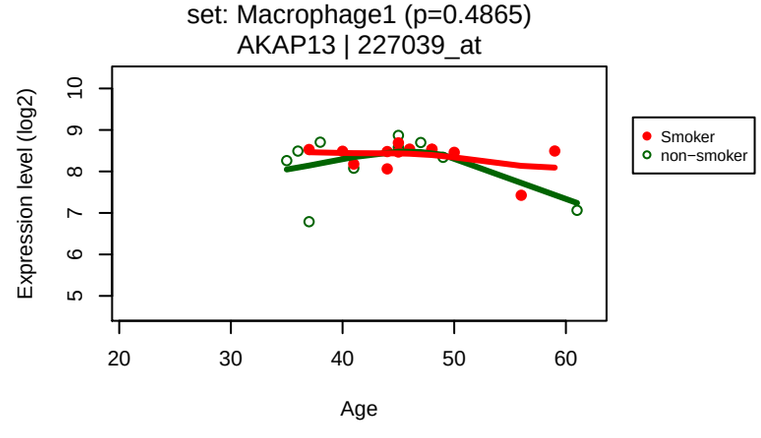

set: SmallAirway1 (p=0.7757)
AKAP13 | 227039_at
Expression level (log2)
Age
Smoker
non−smoker

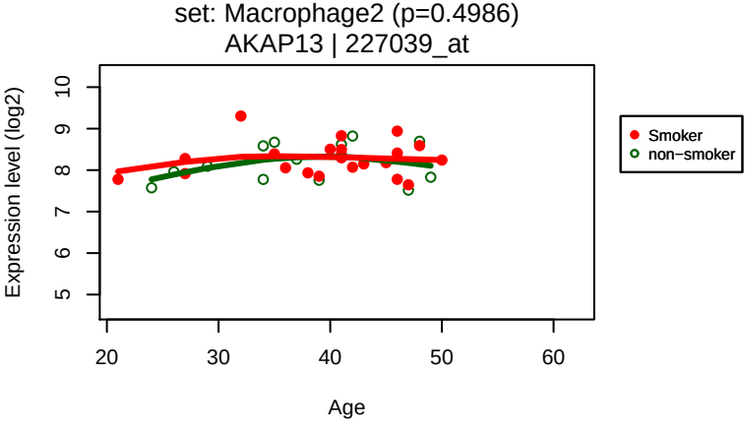

set: SmallAirway2 (p=0.2824)
AKAP13 | 227039_at
Expression level (log2)
Age
Smoker
non−smoker

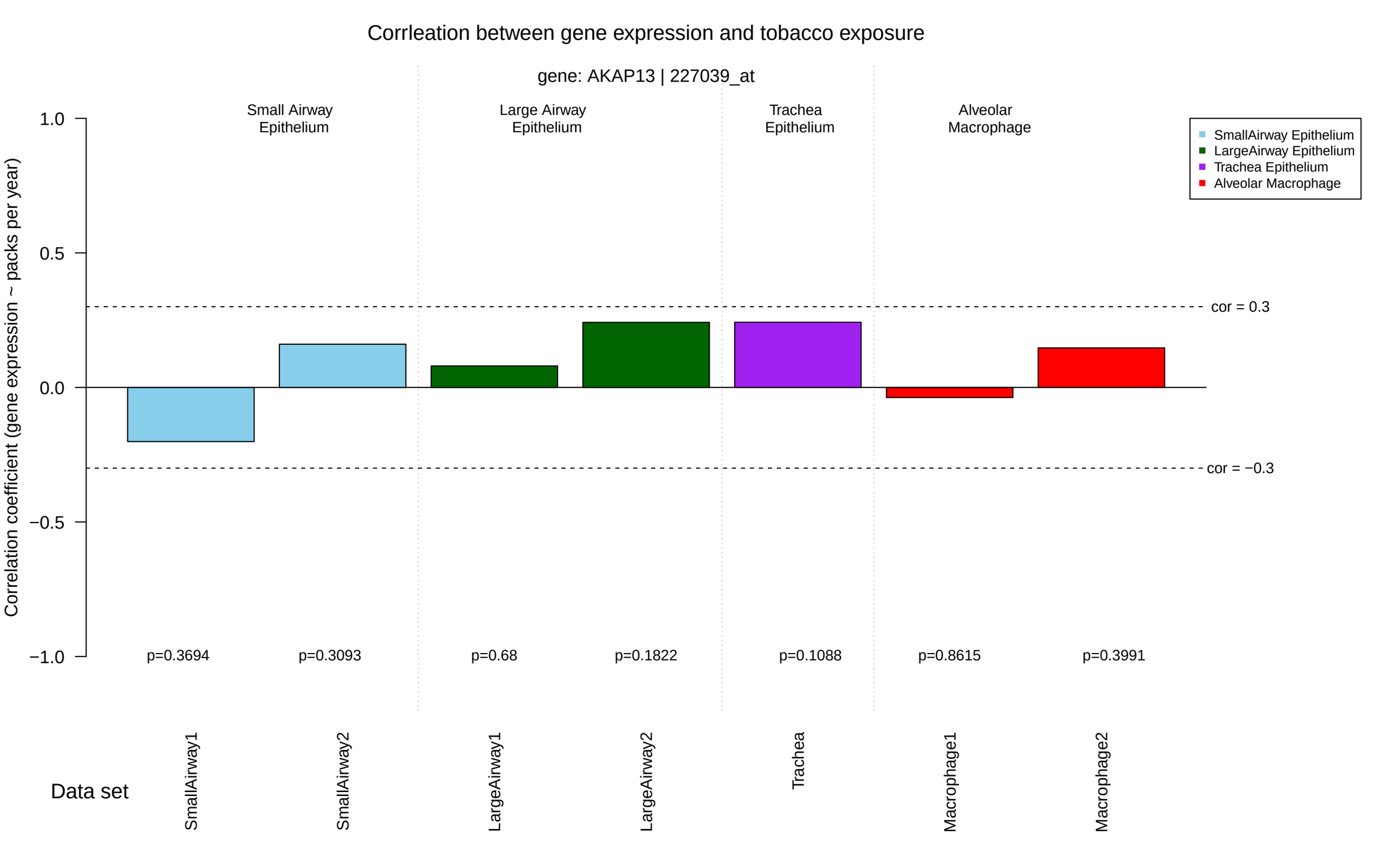

set: LargeAirway1 (p=0.7706)
AKAP13 | 227039_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2613)
AKAP13 | 227039_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0076)
AKAP13 | 227039_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4865)
AKAP13 | 227039_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4986)
AKAP13 | 227039_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AKAP13 | 227039_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3694
p=0.3093
p=0.68
p=0.1822
p=0.1088
p=0.8615
p=0.3991
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2